JOEL THOMAS CYRIL

Electronics and Computer Engineering Applicant | Embedded Systems | Full-Stack Development

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PROFILE

CBSE Class XII student pursuing undergraduate study in Electronics and Computer Engineering, with hands-on experience designing custom electronic devices, developing embedded firmware, and building practical software systems. Projects span ESP32 and STM32 hardware, RF and LoRa communications, robotics, AI-assisted education tools, and self-hosted web applications.

EDUCATION

Abu Dhabi Indian School (ADIS) — Abu Dhabi, UAE

CBSE Senior Secondary (Class XII) • Computer Science, Physics and Mathematics

CBSE Class X board results: **Mathematics 100/100** • **English Language and Literature 95/100**

IIT Madras, Centre for Outreach and Digital Education (CODE) — 2026

Introduction to Electronic Systems • Eight-week certificate course in circuits, semiconductors, op-amps and digital electronics

TECHNICAL SKILLS

Languages	C, C++, Python, TypeScript, JavaScript, SQL, HTML and CSS
Embedded and Control	ESP32, STM32WLE5, FreeRTOS, PID control, sensors
PCB and Interfaces	KiCad, PCB design, I2C, SPI, UART, RS-485
Wireless and RF	LoRa, CC1101 Sub-GHz RF, PN532 NFC/RFID, infrared
Web and Backend	React, Node.js, FastAPI, REST APIs, WebSockets, Vite, Tailwind CSS
Data	PostgreSQL, SQLite, Supabase, Tesseract OCR
Tools and Infrastructure	Linux, Docker, Git/GitHub, Vercel

SELECTED ENGINEERING PROJECTS

Reaper One | Multi-Protocol Handheld Device — 2024–2026

ESP32 • C++ • KiCad • FreeRTOS • CC1101 • NFC/RFID | github.com/JoelCyril/reaper1.2

- Designed and assembled a compact ESP32-based device integrating Sub-GHz radio, NFC/RFID, infrared, Wi-Fi and Bluetooth LE, with a custom circuit board and 3D-designed enclosure.
- Developed menu-driven C++ firmware and integrated modules over SPI, I2C and GPIO; iterated hardware and firmware prototypes over approximately 18 months.

ADIS ReadTrack | AI-Assisted Reading Assessment Platform — 2026

React • TypeScript • Supabase • PostgreSQL • Tesseract OCR

- Developed a school reading and assessment portal in consultation with English teachers, designed to support approximately 2,000 ADIS students.
- Implemented student-card OCR onboarding, reading activity tracking, AI-assisted assessments and class-level literacy leaderboards.

SwapUAE | Cashless Community Marketplace 2026

React • TypeScript • Node.js • PostgreSQL • WebSockets | github.com/JoelCyril/swap

- Co-developed a peer-to-peer barter platform with listings, trade offers and real-time messaging; early launch reached 100+ users, 110+ listings and 12+ trades.
- Built a self-hosted Linux NAS from reused computer parts to support affordable application infrastructure after free cloud limits became restrictive.

Manarat Al-Qurm | Mangrove Monitoring and LoRa Telemetry 2026

STM32WLE5 • LoRa • Raspberry Pi • Python • FastAPI • SQLite

- Designed a low-power coastal monitoring architecture transmitting sensor readings from a buoy to a local Raspberry Pi gateway using LoRa.
- Developed a dashboard for six water-quality parameters and offline 48-hour trend forecasting to support environmental monitoring.

Underwater Remotely Operated Vehicle (ROV) | Robotics 2026

ESP32 • C++ • RS-485 • Brushless motors • PID control

- Developed an underwater ROV using a tethered communication link, brushless thrusters, depth and motion sensors, and PID-based stabilization.

Pulso.ai | AI Wellness Analytics Platform 2026

React • TypeScript • Vite • Data visualization | github.com/JoelCyril/Pulso.ai

- Built an interactive application synthesizing lifestyle inputs into wellness insights and visual dashboards aligned with UN Sustainable Development Goal 3.

Home NAS | Self-Hosted Infrastructure 2026

Linux • PostgreSQL • Docker • ZFS/RAID • Nginx

- Reused discarded computer components to configure local data storage, backups and reverse-proxy services for community applications, reducing dependence on paid hosting.

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Secretary, ADIS Robotics Club 2023–Present

Coordinated workshops and lab activities; mentored junior members in EV3/SPIKE programming and autonomous robotics.

Team Captain, FIRST LEGO League UNEARTHED 2025–2026

Led robot design and mission strategy; team placed First Runner-Up in the regional Robot Game.

Varsity Football, ADIS 2024–2026

Represented the school at the ADEK Cup and the CBSE Cluster U-17 tournament.

Model United Nations Delegate 2024–2025

Participated in committee debate, research and resolution drafting.

SELECTED AWARDS AND RECOGNITION

2026	**RIT Annual Engineering Competition**	First Runner-Up; AED 20,000 scholarship
2026	**ADU STEM Engineering and Coding Competition**	Top 10 of 512 teams (2,083 students, 92 schools)
2026	**Curtin University Dubai STEM Challenge**	Top 12 finalist team, ADIS Elites
2025	**Inter-School English Short Film**	Second position, Indian School Al Ain
2024	**Mayoor Inter-School STEM Competition**	First Runner-Up, senior division
2022–2025	**Chess and Speedcubing**	1st Marthoma Smruthi Chess (2025); 1st ADIS Speedcubing (2022); 2nd School Chess (2024)
2024–2025	**Football and Other Sports**	1st ADIS Inter-House Football (2024); Badminton runner-up (2025); Carrom doubles 2nd (2025)
2023–2024	**Performing Arts**	First prize, Malayalam solo singing